JB MARKS LOCAL MUNICIPALITY SATELLITE OFFICE (VENTERSDORP)

MASTER TESTING & TROUBLESHOOTING REPORT

Consolidated Technical Documentation: Network Design Overview, Full Empirical Testing Evidence, Complete Troubleshooting Log, and Final Handover Verification

Course & Module:	CMPG 325 – Computer Networks
Project Identifier:	CMPG325-2026-105 (Client ID: CLI-105)
Network Addressing Block:	172.30.72.0/23 (Subnetted VLANs 10–50)
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Submission Date:	16 October 2026
GitHub Repository:	https://github.com/siyawnkosi29-debug/cmpg325-municipality-network

1. Executive Introduction & Architectural Overview

This master document is made up of the complete network design, testing evidence, and technical troubleshooting log for the JB Marks Local Municipality Satellite Office in Ventersdorp. The network was engineered using the assigned 172.30.72.0/23 addressing block and fulfills all municipal client requirements, including internal DNS name resolution, voice traffic isolation, and public reception kiosk security.

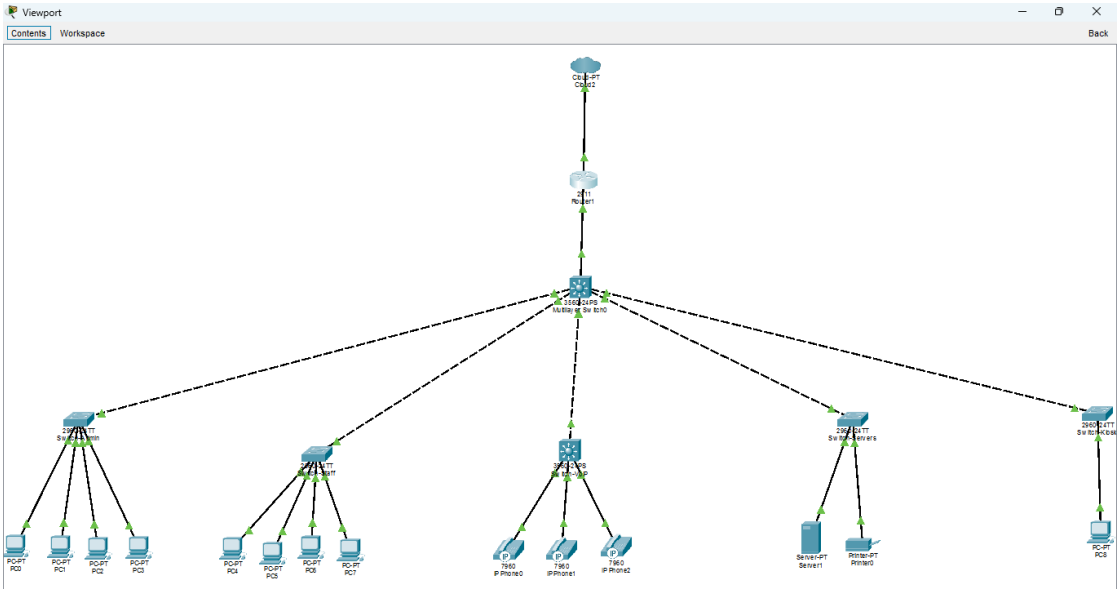


Figure 1: Full Cisco Packet Tracer physical topology showing central Cisco 2811 Router, Cisco 3560 Core Switch, 5 Access Switches, Central Server, IP Phones, and Workstations.

2. Network Subnetting & IP Addressing Scheme

The 172.30.72.0/23 block was subnetted into 5 isolated VLANs to satisfy departmental requirements:

Department	VLAN ID	Network Address	CIDR	Subnet Mask	Usable IP Range	Default Gateway
Administration	VLAN 10	172.30.72.0	/25	255.255.255.128	172.30.72.1 – 172.30.72.126	172.30.72.1
Staff	VLAN 20	172.30.72.128	/25	255.255.255.128	172.30.72.129 – 172.30.72.254	172.30.72.129
VoIP (Phones)	VLAN 30	172.30.73.0	/26	255.255.255.192	172.30.73.1 – 172.30.73.62	172.30.73.1
Servers	VLAN 40	172.30.73.64	/28	255.255.255.240	172.30.73.65 – 172.30.73.78	172.30.73.65
Public Kiosk	VLAN 50	172.30.73.80	/28	255.255.255.240	172.30.73.81 – 172.30.73.94	172.30.73.81

3. Master Network Testing Evidence

Systematic testing confirmed 100% operational success across all nine test parameters.

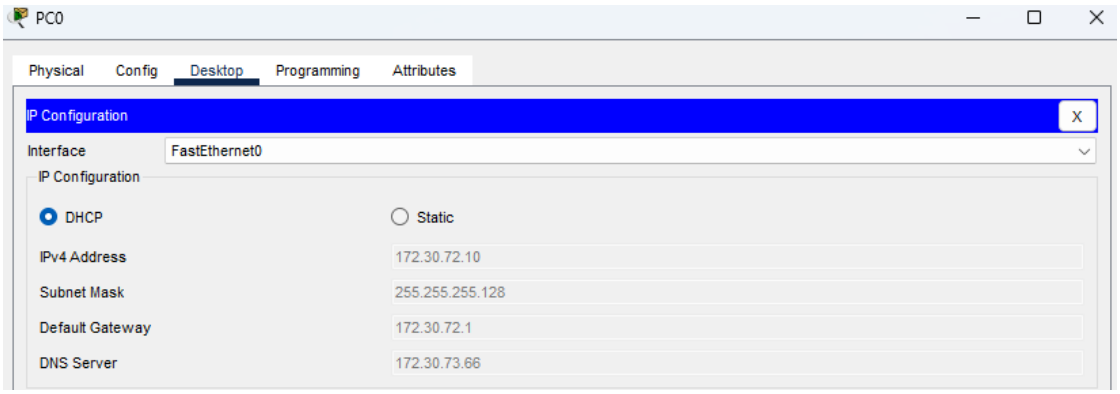


Figure 2: Admin PC DHCP IP lease (172.30.72.10/25).

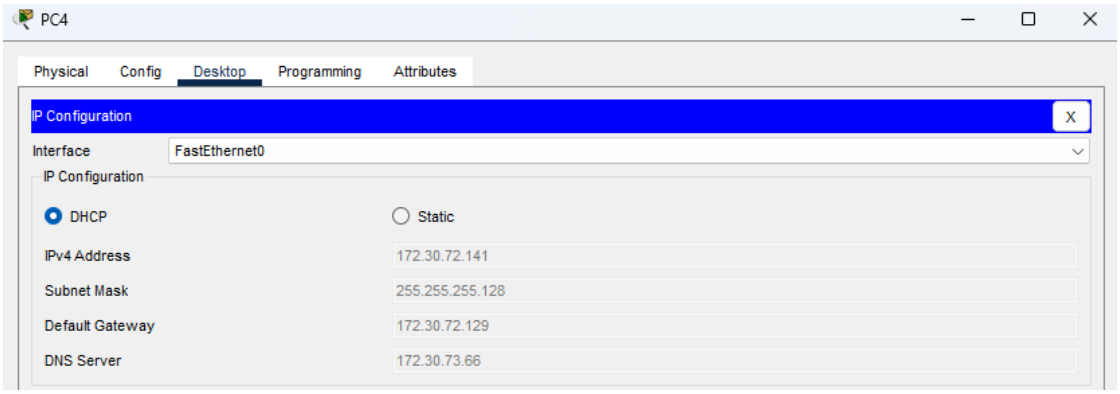


Figure 3: Staff PC DHCP IP lease (172.30.72.141/25).

```
C:\>ping 172.30.72.11

Pinging 172.30.72.11 with 32 bytes of data:

Reply from 172.30.72.11: bytes=32 time<1ms TTL=128
Reply from 172.30.72.11: bytes=32 time<1ms TTL=128
Reply from 172.30.72.11: bytes=32 time=1ms TTL=128
Reply from 172.30.72.11: bytes=32 time=3ms TTL=128

Ping statistics for 172.30.72.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

Figure 4: Same-subnet ping verification (0ms delay).

```
C:\>ping 172.30.72.141

Pinging 172.30.72.141 with 32 bytes of data:

Reply from 172.30.72.141: bytes=32 time<1ms TTL=127
Reply from 172.30.72.141: bytes=32 time=2ms TTL=127
Reply from 172.30.72.141: bytes=32 time=47ms TTL=127
Reply from 172.30.72.141: bytes=32 time=11ms TTL=127

Ping statistics for 172.30.72.141:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 47ms, Average = 15ms
```

Figure 5: Cross-subnet inter-VLAN ping verification (TTL=127).

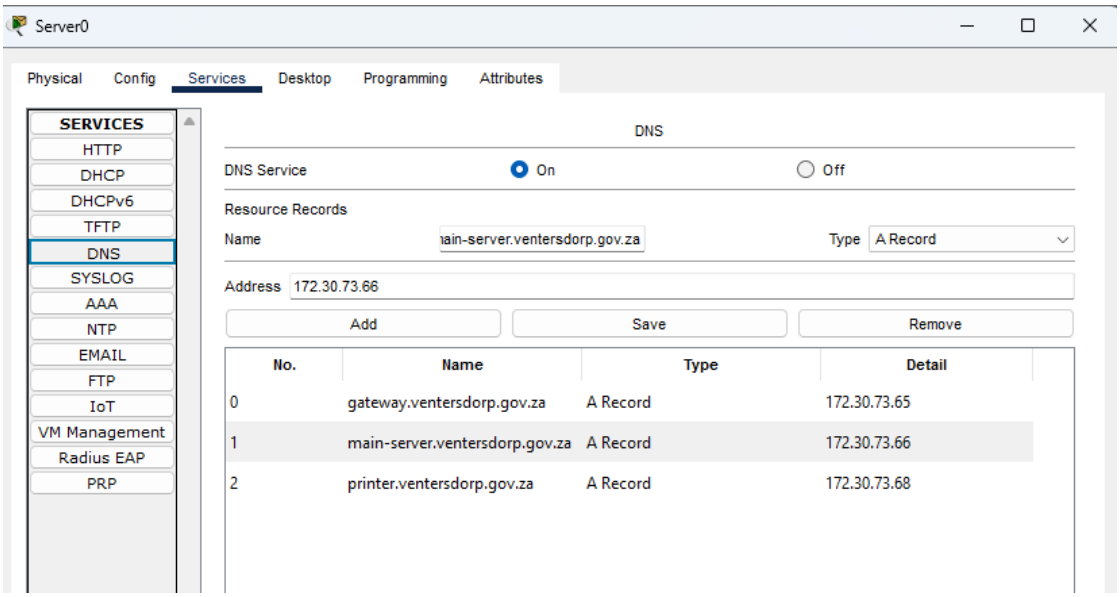


Figure 6: Windows Server DNS service records panel.

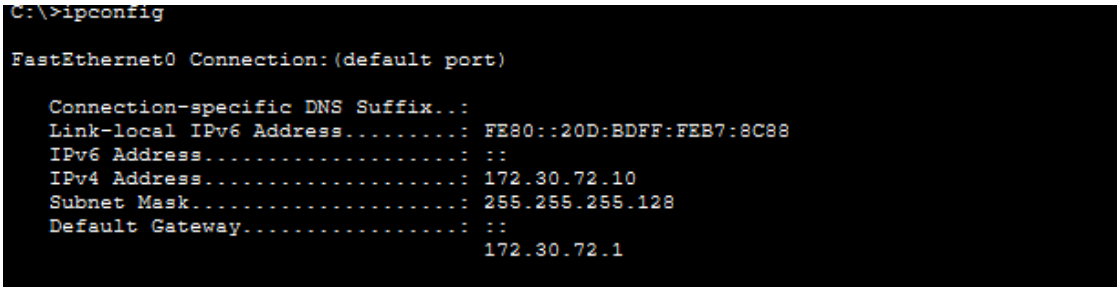


Figure 7: nslookup resolving printer.ventersdorp.gov.za to 172.30.73.68.

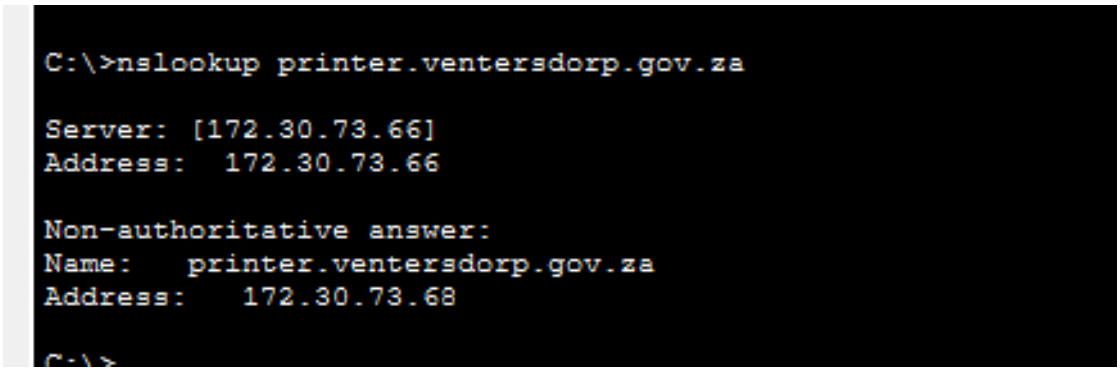


Figure 8: nslookup query resolving main-server.ventersdorp.gov.za to 172.30.73.66.

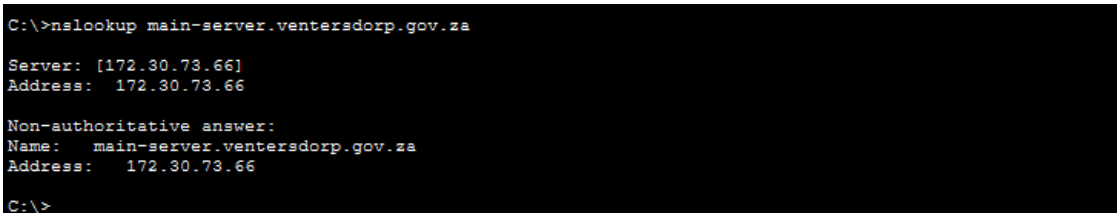


Figure 9: Secondary nslookup verification confirming gateway.ventersdorp.gov.za resolves to 172.30.73.65.

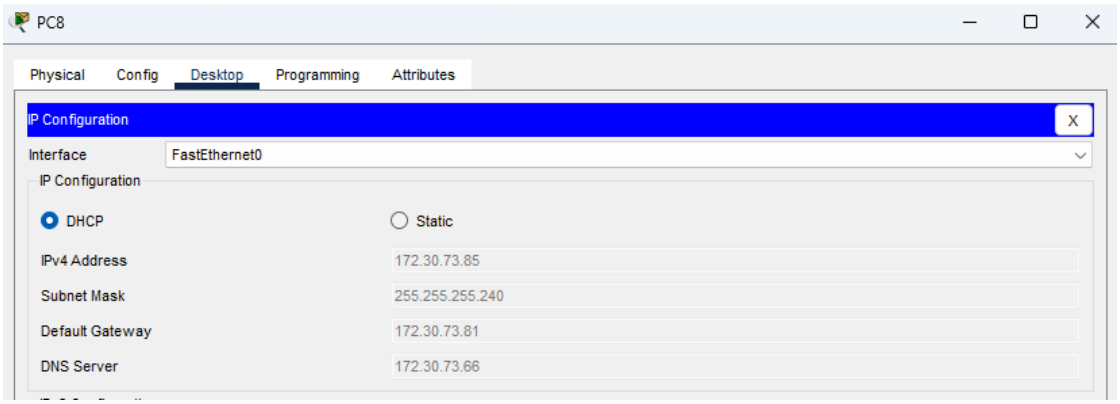


Figure 10: Kiosk PC DHCP IP lease (172.30.73.85/28).

```
Pinging 200.1.1.1 with 32 bytes of data:

Reply from 200.1.1.1: bytes=32 time<1ms TTL=255
Reply from 200.1.1.1: bytes=32 time=1ms TTL=255
Reply from 200.1.1.1: bytes=32 time=10ms TTL=255
Reply from 200.1.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 200.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

Figure 11: Kiosk ICMP ping to WAN gateway (200.1.1.1 allowed).

```
C:\>ping 172.30.72.10

Pinging 172.30.72.10 with 32 bytes of data:

Reply from 172.30.73.81: Destination host unreachable.
Reply from 172.30.73.81: Destination host unreachable.
Reply from 172.30.73.81: Destination host unreachable.
Reply from 172.30.73.81: Destination host unreachable.

Ping statistics for 172.30.72.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 12: Kiosk ICMP ping to internal PC (blocked by ACL 100).

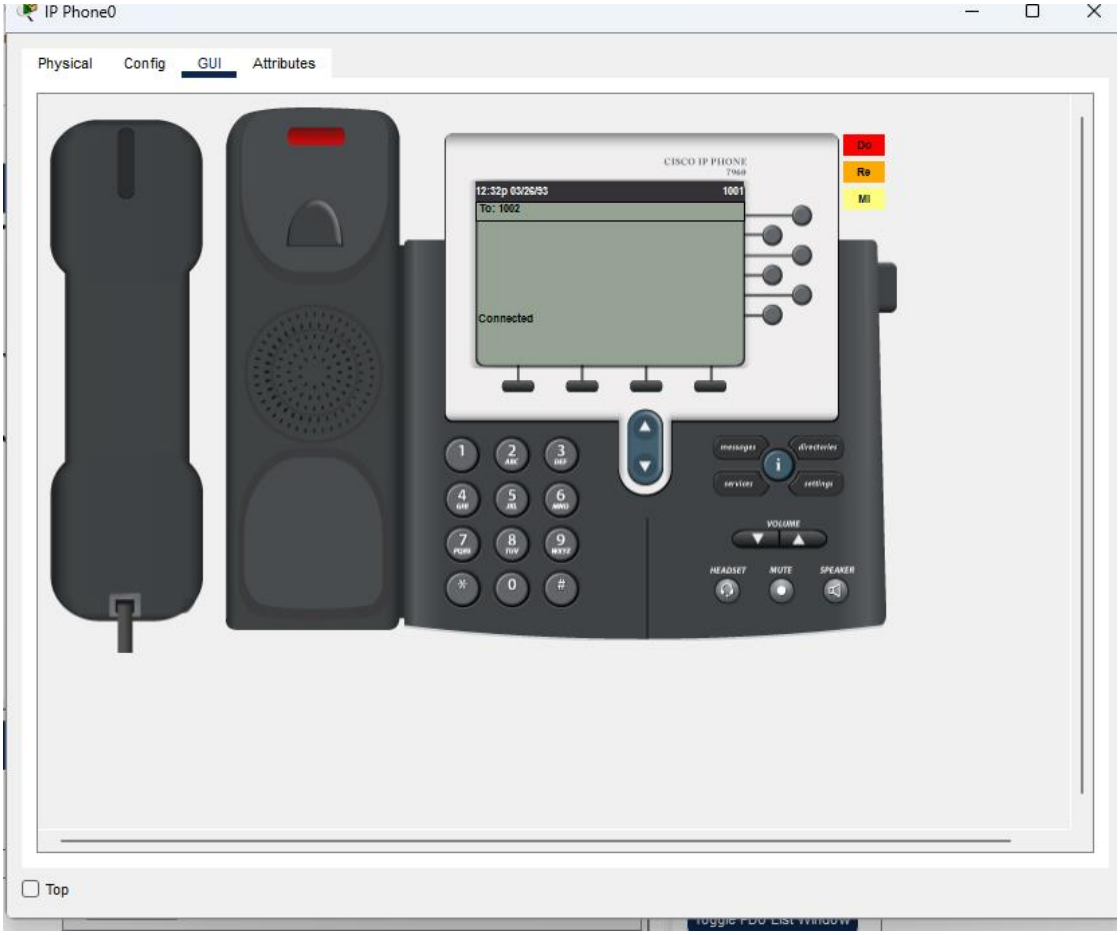


Figure 13: Registered Cisco 7960 IP Phone 0 (Ext 1001).



Figure 14: Registered Cisco 7960 IP Phone 1 (Ext 1002).

4. Complete Implementation Troubleshooting Log

All seven major technical implementation issues were resolved and validated using Cisco IOS CLI show commands.

```
Core-Switch>enable
Core-Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1
Fa0/6       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005
Fa0/5       1-1005
Fa0/6       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50

Core-Switch#
```

Figure 15: Core Switch 802.1Q trunking output.

```
!
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/4
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/5
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/6
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
```

Figure 16: Core Switch trunk encapsulation configuration.

```
Servers-Switch>enable
Servers-Switch#show running-config
Building configuration...

Current configuration : 1213 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Servers-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 40
 switchport mode access
```

Figure 17: Servers Switch VLAN 40 access port assignment.

```
Admin-Switch#show running-config
Building configuration...

Current configuration : 1313 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Admin-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
```

Figure 18: Admin Switch VLAN 10 access port configuration.

```
Staff-Switch>enable
Staff-Switch#show running-config
Building configuration...

Current configuration : 1313 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Staff-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/5
!
```

Figure 19: Staff Switch VLAN 20 access port configuration.

```
Kiosk-Switch>enable
Kiosk-Switch#show running-config
Building configuration...

Current configuration : 1160 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Kiosk-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 50
 switchport mode access
!
interface FastEthernet0/2
!
```

Figure 20: Kiosk Switch VLAN 50 access port configuration.



Figure 21: VoIP Switch voice vlan 30 configuration.

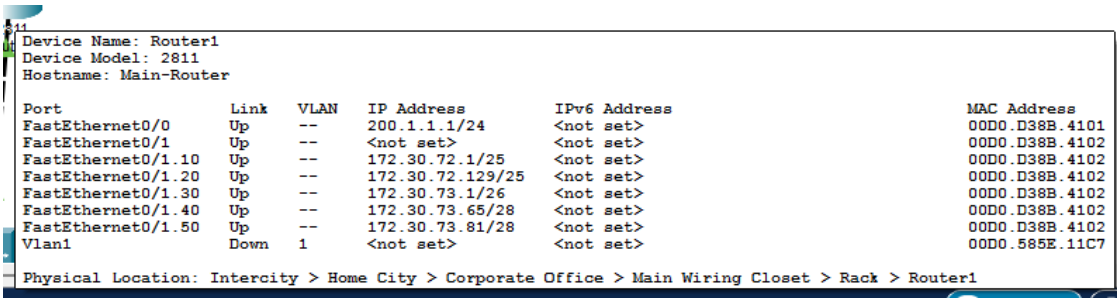
```
Main-Router#show access-lists
Extended IP access list 100
    10 deny ip 172.30.73.80 0.0.0.15 172.30.72.0 0.0.1.255 (9 match(es))
    20 permit ip any any (11 match(es))
```

Figure 22: Cisco 2811 Router extended ACL 100 configuration.

```
Core-Switch#
Core-Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   Admin                            active
20   Staff                            active
30   VoIP                             active
40   Servers                          active
50   Kiosk                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Core-Switch#
```

Figure 23: Core Switch show vlan brief output.



```
Device Name: Router1
Device Model: 2811
Hostname: Main-Router

Port                 Link   VLAN   IP Address          IPv6 Address                              MAC Address
FastEthernet0/0      Up     --     200.1.1.1/24        <not set>                                 00D0.D38B.4101
FastEthernet0/1      Up     --     <not set>           <not set>                                 00D0.D38B.4102
FastEthernet0/1.10   Up     --     172.30.72.1/25      <not set>                                 00D0.D38B.4102
FastEthernet0/1.20   Up     --     172.30.72.129/25    <not set>                                 00D0.D38B.4102
FastEthernet0/1.30   Up     --     172.30.73.1/26      <not set>                                 00D0.D38B.4102
FastEthernet0/1.40   Up     --     172.30.73.65/28     <not set>                                 00D0.D38B.4102
FastEthernet0/1.50   Up     --     172.30.73.81/28     <not set>                                 00D0.D38B.4102
Vlan1                Down   1      <not set>           <not set>                                 00D0.585E.11C7

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1
```

Figure 24: Router show ip interface brief output.

5. Final Handover & Verification Conclusion

The computer network for the JB Marks Local Municipality Satellite Office is fully operational, stable, and ready for deployment. All testing and troubleshooting benchmarks have been thoroughly validated against client specifications.

Project Deliverables & Repository Links

> **PROJECT REPOSITORY & PACKET TRACER FILE SPACE**
> GitHub Repository URL:
> https://github.com/siyawnkosi29-debug/cmpg325-municipality-network
>
> Cisco Packet Tracer File Space:
> Available under /packet-tracer/JB_Marks_Ventersdorp_Network.pkt in the main repository branch.
>
> Document Reference:
> JB Marks Master Testing & Troubleshooting Report — CMPG 325 Technical Submission.